

Introduction of Operating System

1.What is an Operating System (OS)?

-> An Operating System is a software that manages a computer's hardware and allows us to run applications easily.

-> The OS also handles files, controls input and output devices, provides security with permissions and passwords, and allows you to do multiple tasks at the same time—like watching a video while downloading a file.

-> An operating system is a piece of software that manages all the resources of a computer system, both hardware and software, and provides an environment in which the user can execute his/her programs in a convenient and efficient manner by hiding underlying complexity of the hardware and acting as a resource manager.

Example:

Windows, Linux, Android, and iOS are operating systems that help us open apps, save files, browse the internet, and control the device.

2.What happens if there is no Operating System?

-> The computer will not work.

It cannot run any programs, cannot manage memory, and the user cannot control the hardware. The computer becomes useless.

-> Bulky and complex app(Hardware interaction code must be in app's codebase)

-> Resource exploitation by one app.

-> No memory Protection

3.Operating system function?

-> Access to the computer hardware.

-> Interface between the user and the computer hardware

- > Resource management (Aka, Arbitration) (memory, device, file, security, process etc)
- > Hides the underlying complexity of the hardware. (Aka, Abstraction)
- > facilitates execution of application programs by providing isolation and protection.

4.What is System Software?

A: System software controls the computer hardware.

Example: Operating System (Windows, Linux).

5.What is Application Software?

A: Application software helps the user do specific tasks.

Example: Chrome, WhatsApp, MS Word.
